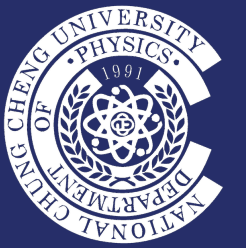


Investigation Influences of Pulling Speed on Stripe Pattern Formation of Peeled Tape



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Introduction

Adhesive tape is commonly used in our daily lives, yet we rarely pay attention to the subtle changes that occur when it is peeled or stretched. In fact, different peeling or stretching conditions can produce a variety of distinct patterns on the tape, which may reflect the underlying physical processes involved in stress and detachment. This study investigates how these patterns change under different pulling conditions, analyzes the mechanisms responsible for their formation, and attempts to identify the underlying patterns.

Experimental Equipment

The experimental setup is shown in (Fig. 1.)

It consists of a stepper motor, a force sensor, and a three-pulley traction system.

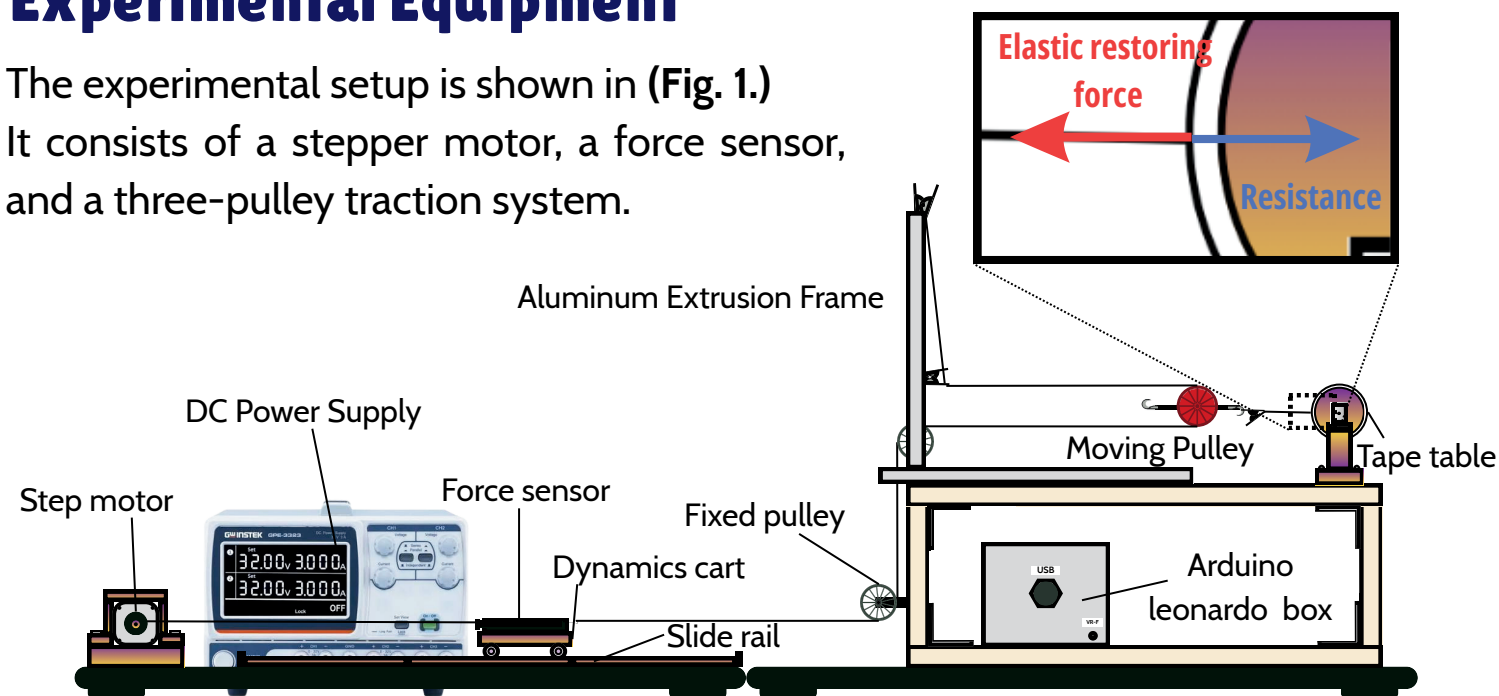


Fig. 1. Schematic diagram of setup for tape peeling experiment

1. Tape Setup: Tape end secured with a metal clamp.
 2. Traction System: Clamp connected to a force sensor and stepper motor via a 3-pulley system.
 3. Speed Control: Stable peeling rate achieved by adjusting the microcontroller
 4. Analysis: Evaluated critical peeling force and stripe spacing at various peeling rates.
- The peeling front is subjected to the competition of two primary forces: the **elastic restoring force of the backing** and the **interfacial adhesive resistance**.

Theoretical Principle

1. Macroscopic Dynamics: Spring-Block Model

Tape peeling is modeled using the classical spring-block model, where the peeled tape acts as an elastic spring and the peeling front as a sliding block. The system follows

$$m\ddot{x} = k(vt - x) - F(\dot{x})$$

where the **elastic restoring force** competes with the velocity-dependent peeling **resistance**.

- **Stick(White Zone):** Elastic energy accumulates while the peeling front remains pinned.
- **Slip(Transparent Zone):** Rapid crack propagation releases the stored elastic energy, producing periodic stick-slip oscillations and stripe patterns.

2. Microscopic Mechanism: Deborah Number and Polymer Viscoelasticity

Pressure-sensitive acrylic adhesive is a viscoelastic polymer whose behavior is governed by the Deborah number (De), defined as $De = \frac{\tau}{t_{obs}}$, where τ is the polymer relaxation time and t_{obs} is the observation time, which is inversely proportional to the pulling speed v .

- **Low-speed peeling ($De < 1$):** Polymer chains have sufficient time to relax, resulting in a rubbery mechanical response. Extensive fibrillation produces the **White Zone**.
- **High-speed peeling ($De > 1$):** Molecular relaxation is restricted, leading to a glassy-like mechanical response. Brittle interfacial fracture with minimal fibrillation produces the **Transparent Zone**.

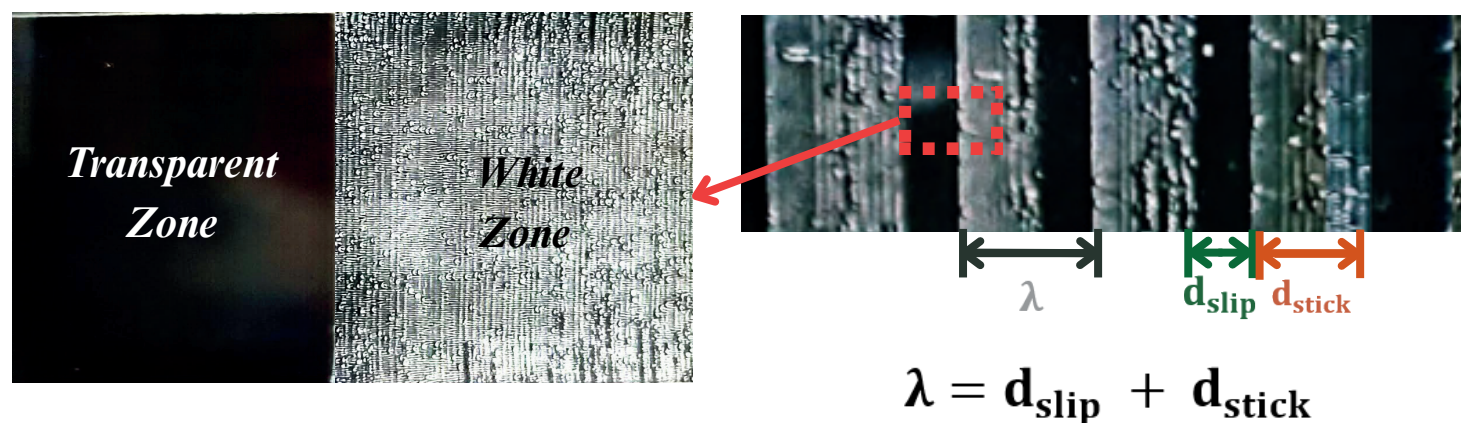


Fig. 2. Characterization of the stripe pattern formation of peeled tape

References

1. M. Grzelka, S. Kooij, S. Woutersen, M. Adda-Bedia, & D. Bonn, "Transition from viscoelastic to fracture-like peeling of pressure-sensitive adhesives," *Soft Matter*, 18(5), 999-1004 (2022).
2. A. I. Everaerts & L. M. Clemens, Chapter 11 - Pressure sensitive adhesives. In D. A. Dillard, A. V. Pocius, & M. Chaudhury (Eds.), *Adhesion Science and Engineering* (pp. 465-534) (2002). Elsevier Science B.V. doi.org

Appendix



Experimental Data,
Code, and Detailed Plots



Interactive Experimental
Setup Simulator



Explore Our Science
Communication Page

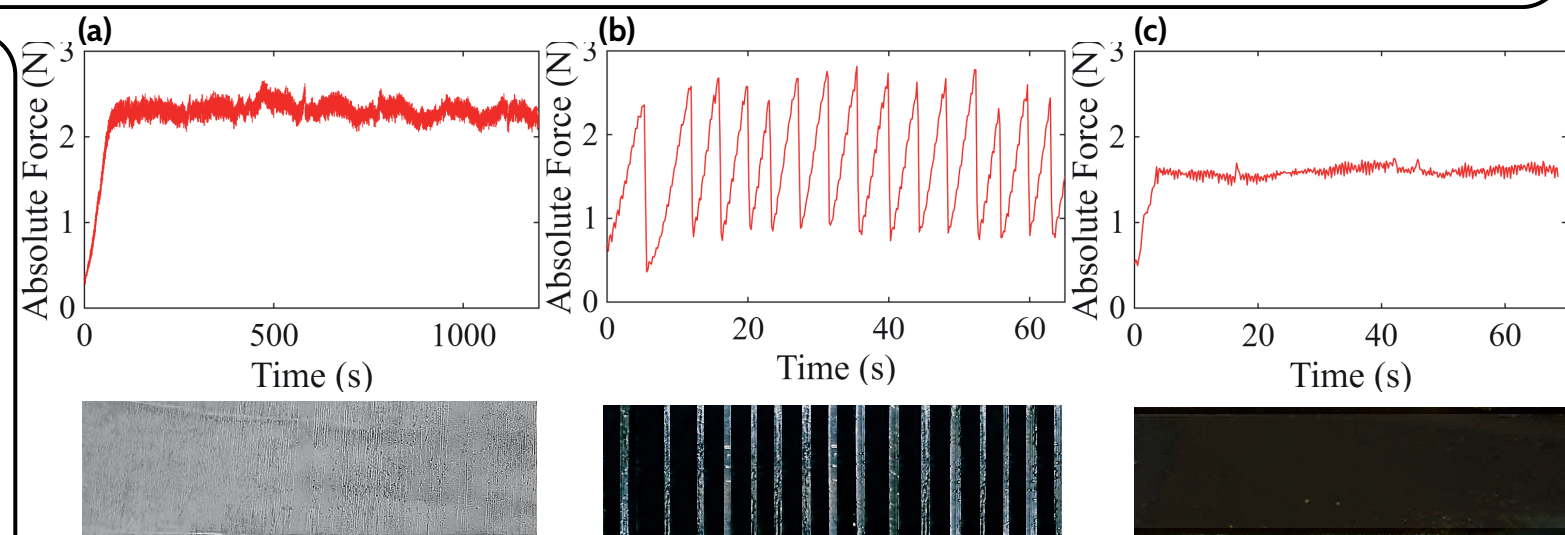


Fig. 3. Time traces of peel-off force and stripe pattern transition VS. pulling speed
(a) Delay 400 (0.115 mm/s) (b) Delay 100 (0.4625 mm/s) (c) Delay 24 (1.927 mm/s)

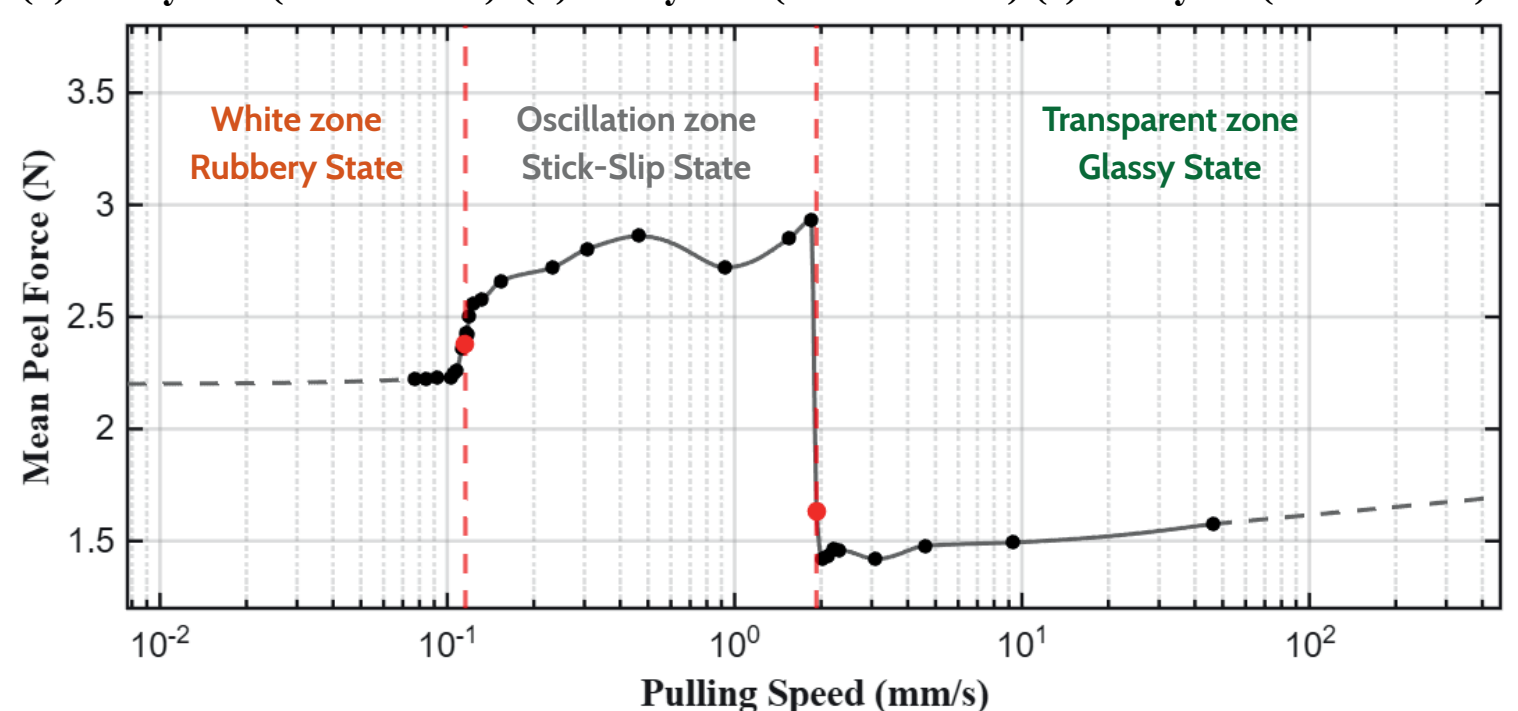


Fig. 4. Dynamic phase diagram of mean peel-off force VS. pulling speed

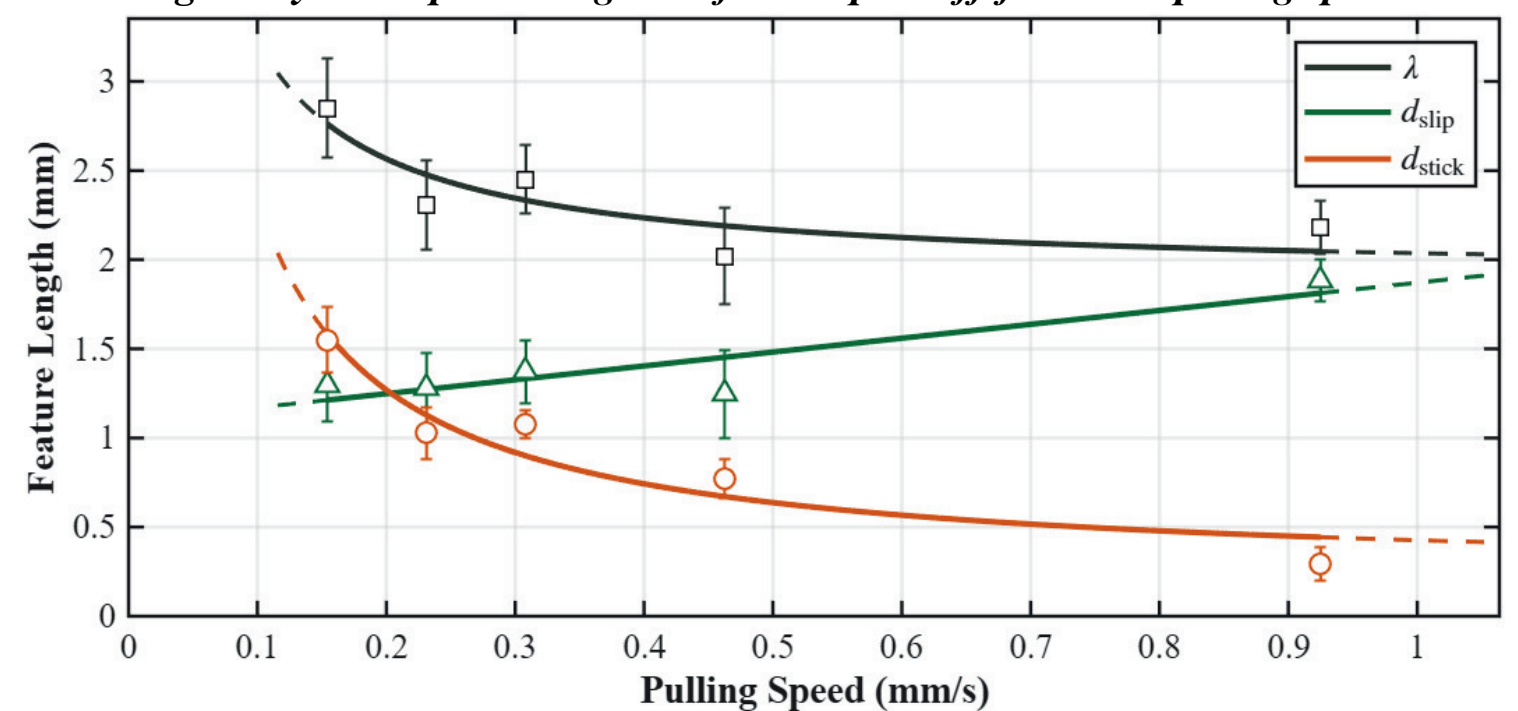


Fig. 5. Characteristic lengths of stripe patterns VS. pulling speed

Summary and Conclusions

1. Manifestation of Three Dynamic Phases (Figs. 3 & 4)

- **Extremely Slow Regime $v < 0.115$ (mm/s) (White zone, Rubbery State):** The polymers have sufficient time to relax, resulting in a steady peeling force that approaches a horizontal plateau.
- **Unstable Regime $0.115 < v < 1.927$ (mm/s) (Oscillation zone, Stick-Slip State):** The system falls into a stick-and-slip cycle, exhibiting violent, sawtooth-like force oscillations.
- **Fast Regime $v > 1.927$ (mm/s) (Transparent zone, Glassy State):** Brittle fracture and viscoelastic stiffening occur, causing the peeling resistance to rise sharply with the pulling speed.

2. Physical Mechanism of Stripe Pattern Formation VS. pulling speed (Fig. 5)

- **White stripe width d_{stick}** are negatively correlated to the pulling speed v : The critical tension is reached rapidly, shortening the fibril drawing zone.
- **Transparent stripe width d_{slip}** are positively correlated to the pulling speed v : Higher sliding kinetic energy and strain rate prolong the slip distance.
- **Stripe spacing λ** : Because the reduction of white stripes dominates, the overall stripe spacing is negatively correlated to the pulling speed, scientifically validating the macroscopic observation that "the faster the tape is peeled, the finer and denser the residual white stripes become."



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